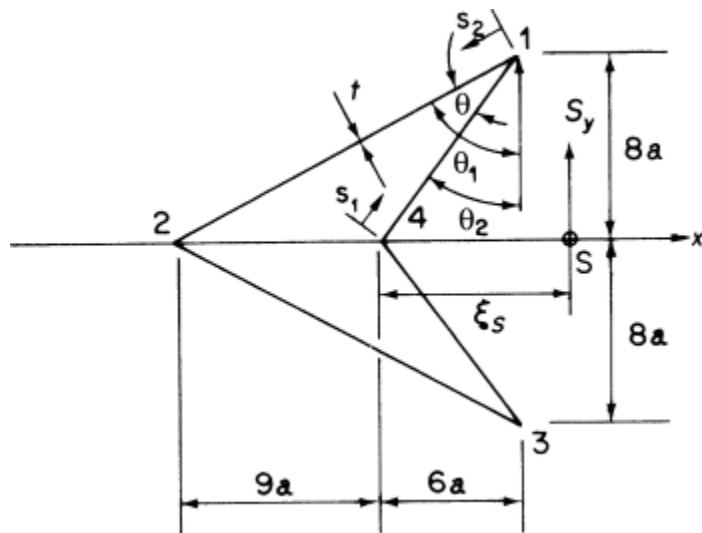


1. Direct stress at a point in the cross-section of a beam subjected to bending depends on
2. The primary assumption made in determining the direct stress distribution produced by pure bending is that Plane cross-sections of the beam remainto the longitudinal fibers of the beam after bending.
3. Plane sections remain plane after bending, assumption is strictly true only if.....are constant along the beam
4. The bending moment M causes the length of the beam to bend about the
5. The shear stresses in a beam are induced by And
6. In symmetrical bending theory if $M_x=0$ then it shows that becomes the neutral axis
7. Although shear flow is constant, shear stress may vary if the is varied.
8. For equilibrium torque must be equal to the generated by shear flow
9. Shear centre and centroid will coincide with each other when the section is
10. Shear centre is the For a section of thin walled beam when subjected to pure shear
11. The magnitude of shear flow is max at the
12. Shear centre is the point about which is zero.
13. Coordinates of the shear centre are property of the cross section
14. Shear stress variation in a beam is
15. The ratio of maximum shear stress to average shear stress in a beam of rectangular section is
16. The ratio of maximum shear stress to average shear stress in a beam of circular section is
17. Greater portion of shear force in a beam of I-section is shared by
18. Shear center in case of a channel section beam will lie
19. Shear stress in a beam is zero at _____

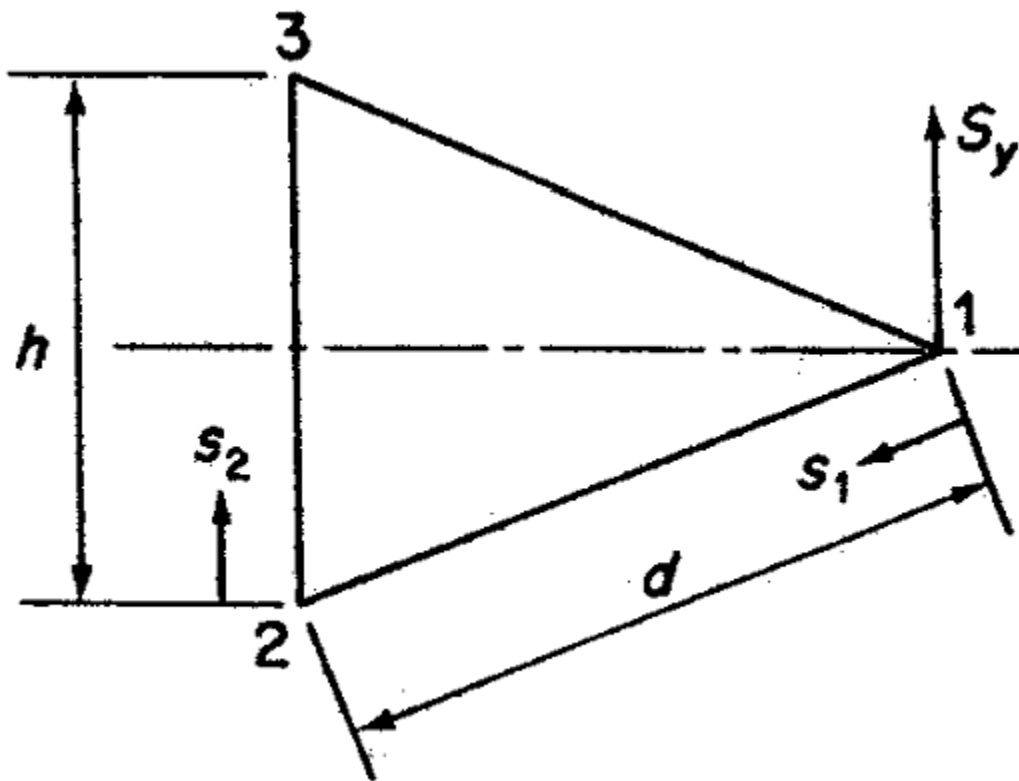
Part B - Main

1. Explain the reason for aero-elastic instabilities with respect to elastic axis and aerodynamic loading.
2. Develop the expression for direct stress for the case of unsymmetrical bending.
3. What are the assumptions of symmetric bending? Explain the unsymmetrical bending with associated equations.
4. Explain shear center in detail
5. Determine the shear flow distribution in the thin walled Z-section.
6. A hollow steel ($G = 11.5 \text{ GPa}$) torsion bar has a circular cross section with outer diameter of 55 mm, inner diameter of 45 mm, and wall thickness t . The mean diameter is 50 mm and $t/D = 0.10$. The length of the bar is 1.50 m. The shear stress at the mean diameter is $\tau = 40 \text{ MPa}$. Determine the applied torque and the angle of twist of one end relative to the other end using the Bredt Batho theory and torsion theory
7. Calculate the distance of the shear centre from the web of the section for a channel cross section
8. Find the value by which the Bredt Batho theory under-estimates the maximum shear stress due to a given torque of uniform hollow tube which has a ratio of inner to outer diameter as 0.9 also determine the error in the angle of twist.
9. A torque $T = 9.50 \text{ kN} \cdot \text{m}$ is applied to the torsion member whose cross section is 60mm 80mm and thickness in the four members are 2,3,4 and 5 mm beginning from the left vertical member. Determine the maximum shear stress in the member.
10. Derive all the equations related to the bredt Batho theorem

11. Calculate the shear flow for the cross section shown in figure



12. Calculate the shear flow for the cross section shown in figure



13. Calculate the shear flow for the cross sections shown below

